



Quantification of Shear Deformations and Corresponding Stresses in the Biaxially Tested Human Myocardium

GERHARD SOMMER,¹ DANIEL CH. HASPINGER,¹ MICHAELA ANDRÄ,² MICHAEL SACHERER,³ CHRISTIAN VIERTLER,⁴ PETER REGITNIG,⁴ and GERHARD A. HOLZAPFEL¹

¹Institute of Biomechanics, Graz University of Technology, Kronesgasse 5/I, 8010 Graz, Austria; ²Division of Cardiac, Thoracic and Vascular Surgery, Klinikum Klagenfurt am Wörthersee, Klagenfurt, Austria; ³Department of Cardiology, Medical University Graz, Graz, Austria; and ⁴Institute of Pathology, Medical University Graz, Graz, Austria

(Received 29 August 2014; accepted 11 February 2015; published online 24 February 2015)

Associate Editor Estefanía Peña oversaw the review of this article.

Abstract—One goal of cardiac research is to perform numerical simulations to describe/reproduce the mechano-electrical function of the human myocardium in health and disease. Such simulations are based on a complex combination of mathematical models describing the passive mechanical behavior of the myocardium and its electrophysiology, i.e., the activation of cardiac muscle cells. The problem in developing adequate constitutive models is the shortage of experimental data suitable for detailed parameter estimation in specific functional forms. A combination of shear and biaxial extension tests with different loading protocols on different specimen orientations is necessary to capture adequately the direction-dependent (orthotropic) response of the myocardium. In most experimental animal studies, where planar biaxial extension tests on the myocardium have been conducted, the generated shear stresses were neither considered nor discussed. Hence, in this study a method is presented which allows the quantification of shear deformations and related stresses. It demonstrates an approach for experimenters as to how the generation of these shear stresses can be minimized during mechanical testing. Experimental results on 14 passive human myocardial specimens, obtained from nine human hearts, show the efficiency of this newly developed method. Moreover, the influence of the clamping technique of the specimen, i.e., the load transmission between the testing device and the tissue, on the stress response is determined by testing an isotropic material (Latex). We identified that the force transmission between the testing device and the specimen by means of hooks and cords does not influence the performed experiments. We further showed that in-plane shear stresses definitely exist in biaxially tested human ventricular myocardium, but can be reduced to a minimum by preparing the specimens in an appropriate manner. Moreover, we showed whether shear stresses can be neglected when performing planar biaxial extension tests on fiber-reinforced materials. The used method appears to be robust to quantify normal and

shear deformations and corresponding stresses in biaxially tested human myocardium. This method can be applied for the mechanical characterization of any fiber-reinforced material using planar biaxial extension tests.

Keywords—Biaxial extension test, Shear deformation, Shear stress, Passive human left ventricular myocardium, Mechanical property.

INTRODUCTION

Most common reasons for mortality are heart related diseases such as diastolic heart failure.¹¹ The pathophysiology of this disease seems to be directly related to the changes of the elastic components in the tissue of the myocardium. In this regard, Yin⁴⁰ stated:

Quantification of ventricular wall stress is necessary for an understanding of both normal and pathological ventricular mechanics. At present there are no reliable means to measure, directly, wall stresses in the intact ventricle. Thus, we must rely on mathematical models to predict these stresses, knowing that the predictions cannot be validated directly.

To better understand how such diseases can occur, it is necessary to obtain more information about the structure and the mechanical behavior, which are closely related to and linked with the electrophysiology of the heart tissue. There is a need to conduct more fundamental research to understand the complex functionality of the human heart. Therefore, one main

Address correspondence to Gerhard Sommer, Institute of Biomechanics, Graz University of Technology, Kronesgasse 5/I, 8010 Graz, Austria. Electronic mail: sommer@tugraz.at

goal of cardiac research is to create numerical (finite element) simulations in order to describe and reproduce the (mechano-electrical) function of the human myocardium in health and disease.^{5,6} By means of computer simulations, new methods of treatment could be developed and tested, which, for example, could enable the function of the human heart to be rebuilt after a myocardial infarction. Numerical simulations are based on material models, which have to be established according to the complex structure and the nonlinear mechanical properties of human heart tissues.

Shear tests in different directions of porcine hearts by Dokos *et al.*³ have shown that the myocardium has orthotropic mechanical properties and, therefore, cannot be *fully* characterized using biaxial extension tests.¹⁵ It is suggested that the combination of data sets from multiaxial mechanical tests, i.e., planar biaxial extension and triaxial shear tests, is able to adequately capture the direction-dependent (orthotropic) material behavior.¹⁵ Multiaxial mechanical tests mimic the physiological loadings better than uniaxial tests and, therefore, provide direction-dependent mechanical information of the tested tissues. For example, biaxial test data with different loading protocols are able to simulate similar loading conditions as the ventricular myocardium experiences *in vivo* during the relaxation and/or filling phases (atrial systole). From biaxial test data obtained from different loading protocols and shear test data at different specimen orientations more sophisticated parameters for constitutive models of the myocardium can be determined.¹⁵ By coupling a mechanical constitutive model for the passive state with an electrophysiological model, which describes the activation of the cardiac muscle cells, the required numerical model can then be obtained providing a better understanding of fundamental mechanisms of the complex mechano-electrical features of the heart walls. Presently, to the authors' knowledge there are no biaxial extension or triaxial shear data available concerning the human myocardium. Nevertheless, this study only focuses on the performance and optimization of planar biaxial extension tests on thin slaps of the human ventricular myocardium, a necessary preliminary study for a more comprehensive biomechanical study where biaxial extension data, triaxial shear data, and microstructural data of passive human myocardium are combined.³⁵

In order to characterize soft biological tissues in the field of biomechanics as accurately as possible, it is of utmost importance to also study the mechanical behavior and its causal morphology within a multiscale approach, i.e., from the microscale to the macroscale level. The human heart is a muscular, hollow organ, and its wall can be differentiated into three compo-

nents: (i) an outermost connective tissue layer (epicardium), (ii) a medial muscular layer (myocardium), (iii) an innermost endothelial layer (endocardium). The epicardium and the endocardium are approximately 100 μm thick membranes consisting of connective tissues such as collagen, elastin and endothelial cells, respectively. Previous studies by, e.g., Frank and Langer,⁷ Young *et al.*⁴² and Sands *et al.*²⁹ have shown that the myocardium consists mainly of myocytes (approximately 70% of the volume) and connective tissue arranged in parallel. About 2 to 5% of the connective tissue is collagen (type I and III), which is organized in a network that surrounds muscle fibers and gives the ventricle its passive mechanical properties. The organization of muscle fibers in the myocardium was described by different methods such as myofiber angle measurements on histological sections,³⁷ confocal microscopy²⁸ or Diffusion Tensor Magnetic Resonance Imaging (DT-MRI).^{27,30} It is established that muscle fibers are arranged almost in parallel and form a laminar structure, so-called sheets, of three to four muscle fibers in the thickness transversely oriented to the heart wall. Furthermore, analysis of these investigations has shown that the fiber orientation can be described by a fiber angle α , which changes continuously through the wall thickness, namely from approx. -60° epicardially to $+60^\circ$ endocardially in the human LV myocardium.²⁷ Furthermore, a collagen network that coats every single muscle fiber and connects neighboring fibers together, called the endomysium, was characterized. In addition to the endomysium, there is also the perimysium, which is oriented towards the sheets.

Hence, by performing biaxial extension tests on prepared tissue, slices from different depths (see Fig. 1) would theoretically result in different passive mechanical behaviors based on the rotation of the fiber angles through the wall thickness of the heart.¹⁹ Due to this particular fiber distribution, it is necessary to investigate if there are also shear stresses present in addition to (typical) normal stresses. If shear stresses are neglected then it is assumed that the calculated normal stresses are automatically principal stresses, which is, in general, not the case.

Neglecting shear stresses may lead to systematic errors in the material parameter quantification. In most published papers, where planar biaxial extension tests on animal hearts have been conducted, influences based on shear were neither considered nor discussed.^{1,2,10,24,25} Therefore, the goal of this preliminary study is to determine occurring normal and shear deformations and their corresponding stresses during biaxial extension tests of the human myocardium, and how they are influenced by different preparation methods. In other words, this study shows whether

shear stresses can be neglected when performing planar biaxial extension tests in contrast to previous biaxial studies.^{1,2,10,24,25} Moreover, the method presented herein allows the quantification of shear deformations and related stresses of several biaxially tested soft biological tissues.

MATERIALS AND METHODS

Material

In this study, myocardial tissue specimens ($n = 14$) from the left ventricles of nine human hearts (mean age 65 ± 7) were obtained from the Clinical Department of Transplant Surgery and the Institute of Pathology, Medical University of Graz, Austria. The use of myocardial material from human subjects for this study was approved by the Ethics Committee, Medical University of Graz, Austria (no. 24-107 ex 11/12 and no. 24-003 ex 11/12). As soon as the Clinical Department of Transplant Surgery declares a heart as non-transplantable, it was explanted, dissected, and appropriate parts of the left ventricle were stored in a cardioplegic solution with BDM (2,3-Butanedione monoxime) in order to prevent the muscle cells from activating and, therefore, to conserve nutrients and cells. Heart tissue originating from the Institute of Pathology is not provided with oxygen and nutrients after the patient's death and, therefore, autolysis

occurs. Thus, this tissue becomes unusable for the determination of representative mechanical behavior if these processes have advanced for too long (> 24 h). To prevent incorrect experimental data in both cases, it is very important that the cold chain is not interrupted and experiments are performed as fast as possible after explantation. Therefore, the tissue is stored in a cardioplegic solution at approximately 4°C between explantation and preparation and tested within 24h after death.

Specimen Preparation

Testing a soft biological tissue, especially the human myocardium, poses challenges to the experimenter, starting from the preparation and the clamping of the tissue through the performance of experiments. The structure and composition of the human myocardium mostly influence the passive mechanical properties of the tissue. Components of the myocytes and the extracellular matrix, such as endomysial and perimysial collagen, are responsible for that behavior. The displacement field of the equibiaxially tested human myocardium largely depends on the orientation of the myofibers in the tissue, and, therefore, the general three-dimensional deformation is composed of normal and shear deformations.

We started the specimen preparation by cutting a thin slice (≈ 0.5 mm) from the outside to remove the

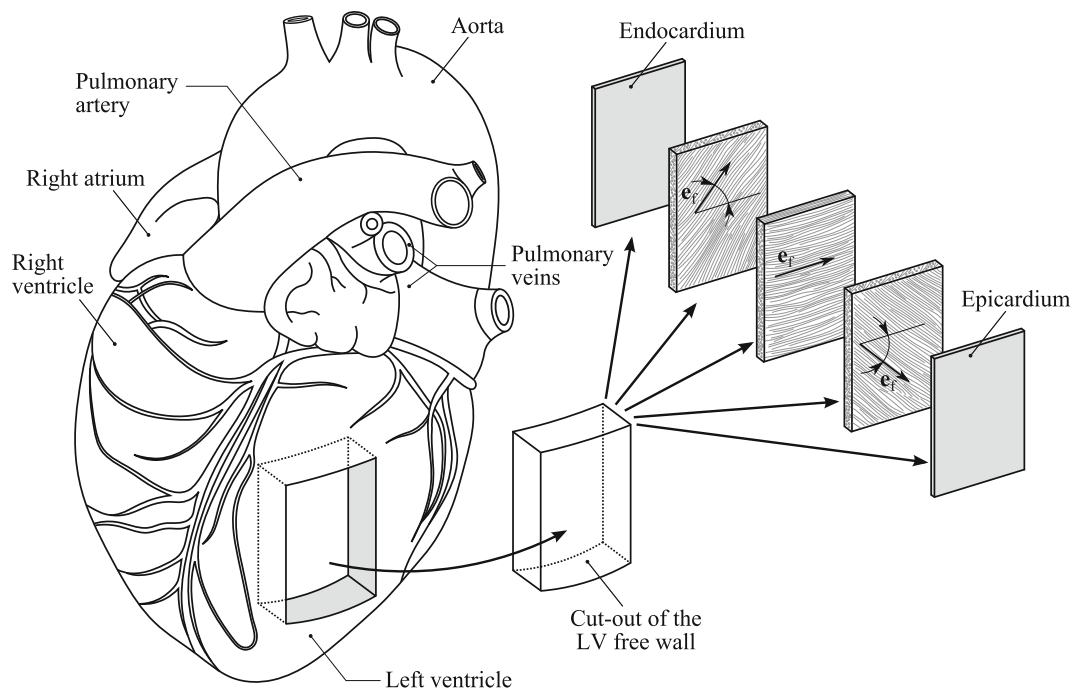


FIGURE 1. Schematic diagram of the excision of biaxial specimens from the left ventricle of a heart. The myocardium is separated from the endocardium and the epicardium, and the myocardium is dissected into slices with the same thickness. The orientation e_f of muscle fibers changes through the thickness of the myocardium.

epicardium with the aid of a meat slicing machine. With the same machine, the myocardium could then be subdivided into several thin slices with an average thickness of 2.3 ± 0.5 mm ($n = 14$), see Fig. 1. Since we can only track in-plane deformations of the biaxially tested specimen, we prepared specimens as thinly as possible in order to reduce the influence of the ‘out of plane’ shearing during testing. We identified the dominant orientation of muscle fibers on the upper and lower surface of each slice, and determined an average fiber direction of the continuum, say $\mathbf{e}_f$, which we subsequently call the mean fiber direction (MFD). Due to the laminar structure of the myocardium, the determination of the fiber orientations was even possible with the naked eye. The angle that occurs between the MFD and the fiber orientation $\mathbf{M}$ of the upper surface of the specimen, say α , is assumed to be the same as the angle between the MFD and the fiber orientation $\mathbf{M}'$ of the lower surface, as shown in Fig. 2.

The illustrated specimen is embedded in a reference frame with coordinate axes (x_1, x_2) and a fixed set of orthonormal basis vectors $(\mathbf{e}_1, \mathbf{e}_2)$. The angle between the basis vector $\mathbf{e}_1$ and the mean fiber direction characterized by $\mathbf{e}_f$ is identified as β . For the special cases $\beta = 0^\circ$ or 90° , the (average) in-plane shear deformations are zero by stretching the specimen along (and orthogonal to) the MFD, due to the ‘symmetry’ of $\mathbf{M}$ and $\mathbf{M}'$. Of course, it is still assumed that the fiber orientation does change continuously through the thickness of the specimen. From a theoretical point of view, shear deformations definitely exist within the specimen for other values of β between 0° and 90° . In Fig. 2, the recorded forces in the $\mathbf{e}_1$ - and $\mathbf{e}_2$ -directions are indicated as f_1 and f_2 , respectively.

As mentioned above, for each slice a MFD was determined. Since we want to quantify the influences of different preparation methods on the mechanical properties of the human myocardium, we designed two distinct fiber-alignment cases. By using a template, a block with a square cross section (approx. 25×25 mm) was cut out from the slice so that the angle between the MFD, $\mathbf{e}_f$, and the basis vector $\mathbf{e}_1$ was $\beta = 0^\circ$ (testing mode 1, see Fig. 3a) for one group of samples ($n = 7$), and $\beta = 45^\circ$ (testing mode 2) for the other group ($n = 7$). The used template was based on the findings of Eilaghi *et al.*,⁴ and was especially developed for these experiments. Eilaghi *et al.*⁴ studied the design of specimens and boundary conditions concerning the load transmission using hooks and cords. Their finite element simulations suggested that in order to obtain the best results, five hooks per side with the same distance apart from one another should be used, and the edge of the specimen should not be

more than 0.6 to 1.0 times the distance between two hooks. Furthermore, they discovered greater influences of the clamping method when specimen dimensions were smaller.

Nine markers made of black foam rubber were fixed as close to the center as possible using a very thin layer of cyanoacrylate gel. Histological investigations of the surface (not shown in this paper) indicated that the adhesive penetrated the tissue surface marginally ($\approx 50 \mu\text{m}$). These markers were fixed onto the surface of the specimen at a distance of approximately 2.5 mm from each other (for tracking the deformation of the tissue during tests), and hooks connected to surgical sutures were positioned (for load transmission between tissue and testing machine), as can be seen in Fig. 3b. The surgical sutures are connected to specially designed carriages that allow self-equilibrated loads of each suture line.

To show the efficiency of the above described preparation techniques, we quantified in-plane shear stresses and compared them to calculated normal stresses. In addition, shear stresses may also result from poor load transmission. To quantify that effect, an isotropic elastic material (Latex) was also tested.

Biaxial Tensile Test

The used planar biaxial extension testing machine was designed specifically for the mechanical characterization of soft biological and artificial materials.^{33,34} This setup integrates four linear actuators which can be controlled independently by position, force or stretch. The force measurement is conducted by submersible load cells: two in the direction $\mathbf{e}_1$ and two in the direction $\mathbf{e}_2$ (see Fig. 2), which are connected to a clamping device. The maximum testing load is specified at 100 N per sensor and achieves a resolution of 0.6 mN. The maximum travel range is 50 mm (with a resolution of $0.1 \mu\text{m}$) for each linear actuator, and the maximum speed is limited to approximately 30 mm/s. Another feature of this biaxial extension testing device is the video-extensometer (VE), which allows two-dimensional, non-contact stretch measurements by tracking markers which are fixed onto the surface of the specimen. A resolution of $0.15 \mu\text{m}$ of the VE is specified by the manufacturer.

In this study, stretch-controlled tests were performed by using the following test protocol:

- (i) Tensile tests at different subsequently applied stretch levels were performed: $\lambda = 1.05, 1.075, 1.10, 1.125, 1.15, 1.175$, and 1.20 (according to the elasticity of the tissue).

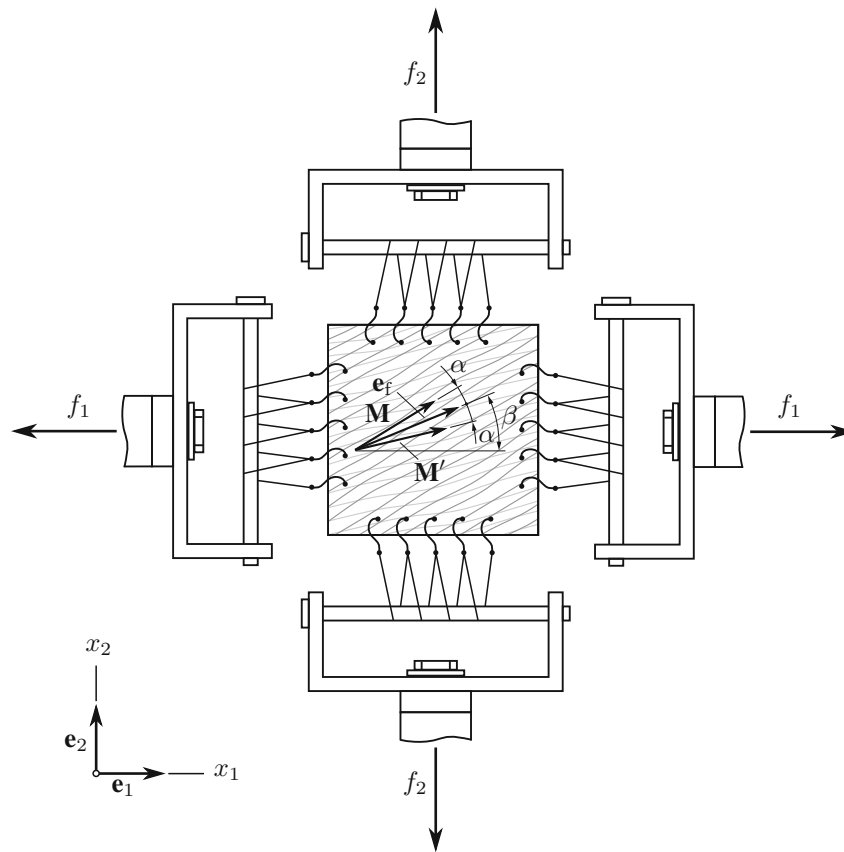


FIGURE 2. Schematic diagram of the biaxial extension test setup: M denotes the fiber orientation on the upper surface of the sample, while M' stands for the fiber orientation on the lower surface; e_f points in the averaged fiber direction, the so-called mean fiber direction of the continuum body, where α corresponds to the angle between e_f and M , and e_f and M' , respectively. The angle β , which occurs between the basis vector e_1 and e_f , describes the in-plane orientation of the mean fiber direction within the specimen.

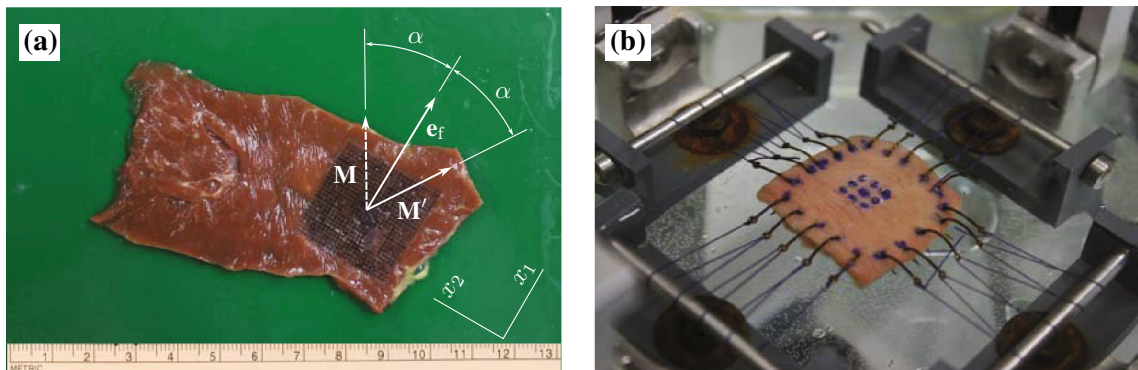


FIGURE 3. (a) Preparation of a myocardial slice for biaxial extension tests in testing mode 1. After determining the muscle fiber orientation on the upper and lower surface (M , M'), a template is positioned in such a way that the mean fiber direction e_f is aligned with the coordinate axis x_1 . (b) Prepared specimen inserted into the testing device, ready for testing.

- (ii) Four preconditioning cycles and one 'measurement' cycle were executed, in order to get a reproducible response (Fig. 7 in the results section shows that four conditioning cycles were enough).

- (iii) Testing speed: 3 mm/min (quasi-static).
- (iv) During testing, specimens were immersed in cardioplegic solution at 37°C. Specimens from the Clinical Department of Transplant Surgery were additionally treated with BDM

(30 mmol/l) to ensure mechanically passiveness of the myocardium.

Data Analysis

Particle Tracking

To control the biaxial extension test, in-plane deformations of the specimen are recorded by means of a CCD camera. The tracking software ‘Laser Speckle Extensometer’ (LSE) evaluates the displacement of the markers fixed onto the specimen surface in real time. In addition to ‘normal strains’ along the x_1 - and x_2 -axes, shear deformations were also determined. An existing tracking algorithm for Matlab, which was used by Sharpe *et al.*,³¹ was adapted for this task. This code is based on ‘Differential Digital Image Tracking’ (DDIT) techniques, which have a much greater resolution (subpixel) than conventionally used ‘Digital Image Correlation’ techniques, with a maximum resolution of one pixel. In contrast to the LSE software, it is now possible to determine and save actual pixel coordinates of markers with the DDIT technique. Therefore, a video (30 frames per second) of the testing procedures was recorded and later subdivided into single images (every fifth image was considered). The DDIT technique used finds actual marker positions based on an intensity distribution of a picture. A black marker (dark) provides a relatively sharp contrast with respect to the background of the tissue (bright). A marker has a peak in the intensity distribution, both in the longitudinal and the transverse direction, which can be approximated to a normal distribution. Hence, a Gaussian distribution is fitted to the real intensity distribution for each marker in both directions by using a nonlinear least-squares method. These maxima provide pixel coordinates of the intensity peaks, and actual marker positions can be calculated in terms of millimeters by means of a reference scale. This process was performed for each image, and actual marker position coordinates were saved.

Kinematics—Measure of Deformation

From the recorded marker positions, we can determine a displacement field within the tracked area and calculate existing strains by performing partial derivatives.¹⁷ In order to describe the whole deformation field, which is bounded by the markers 1 to 4 (see Fig. 4), deformations in between the markers also have to be determined. By means of isoparametric interpolations, a relation between the measured displacements of each marker and the displacements of an arbitrary

point within the area of interest (square, restricted by the markers 1 to 4) has to be found.

A common strategy is to transform the measured marker positions ($X_1^{(n)}, X_2^{(n)}$) in the reference configuration (marker labeling is $n = \{1, \dots, 9\}$) into a coordinate system (r, s) by means of interpolation functions, so-called shape functions, i.e.

$$X_1(r, s) = \sum_{n=1}^9 h^{(n)}(r, s) X_1^{(n)}, \quad X_2(r, s) = \sum_{n=1}^9 h^{(n)}(r, s) X_2^{(n)}, \quad (1)$$

where $h^{(n)}$ are shape functions and r and s are parameters that vary according to their locations n ($-1 \leq r \leq 1, -1 \leq s \leq 1$). The shape functions have to be chosen so that they are equal to one at the marker n , and zero at all the other markers, i.e.

$$\begin{aligned} h^{(1)}(r, s) &= 1/4(1+r)(1+s) - 1/2h^{(5)} - 1/2h^{(8)} - 1/4h^{(9)}, \\ h^{(2)}(r, s) &= 1/4(1-r)(1+s) - 1/2h^{(5)} - 1/2h^{(6)} - 1/4h^{(9)}, \\ h^{(3)}(r, s) &= 1/4(1-r)(1-s) - 1/2h^{(6)} - 1/2h^{(7)} - 1/4h^{(9)}, \\ h^{(4)}(r, s) &= 1/4(1+r)(1-s) - 1/2h^{(7)} - 1/2h^{(8)} - 1/4h^{(9)}, \\ h^{(5)}(r, s) &= 1/2(1-r^2)(1+s) - 1/2h^{(9)}, \\ h^{(6)}(r, s) &= 1/2(1-s^2)(1-r) - 1/2h^{(9)}, \\ h^{(7)}(r, s) &= 1/2(1-r^2)(1-s) - 1/2h^{(9)}, \\ h^{(8)}(r, s) &= 1/2(1-s^2)(1+r) - 1/2h^{(9)}, \\ h^{(9)}(r, s) &= (1-r^2)(1-s^2). \end{aligned} \quad (2)$$

In order to reach a high precision for this interpolation, the original marker locations 1 to 4 should be placed at the corners of a quadrilateral. The markers 5 to 8 should be positioned preferably in the middle, between the endpoints of each side, and the marker 9 should be in the center of the constructed square. For further details the reader is referred to, e.g., Zienkiewicz and Taylor.⁴³

Now, the displacements can also be described by means of this isoparametric formulation, as a function of r and s , where $u_1^{(n)}$ and $u_2^{(n)}$ are the measured displacements along the two coordinate axes, and denote the interpolated displacements of an arbitrary point within the nodes 1 to 4, i.e.:

$$u_1(r, s) = \sum_{n=1}^9 h^{(n)}(r, s) u_1^{(n)}, \quad u_2(r, s) = \sum_{n=1}^9 h^{(n)}(r, s) u_2^{(n)}. \quad (3)$$

With this formulation for the displacement $[\mathbf{u}] = [u_1(r, s), u_2(r, s)]^T$ and the geometry of an element in the reference configuration $[\mathbf{X}] = [X_1(r, s), X_2(r, s)]^T$, it is now straightforward to calculate partial deriva-

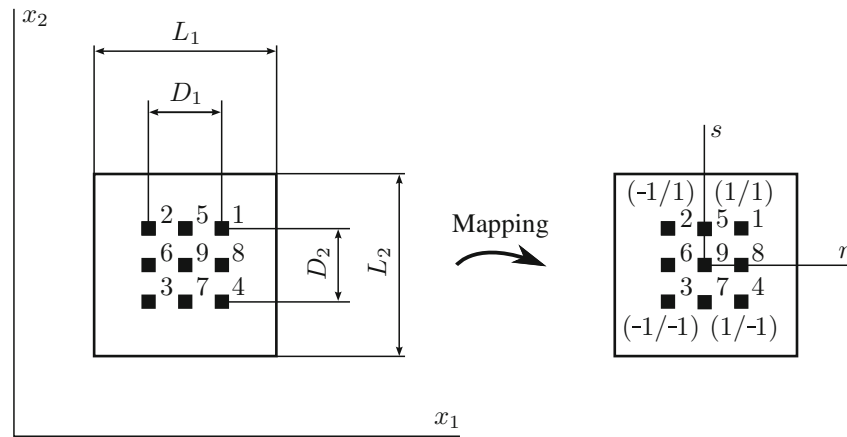


FIGURE 4. Schematic diagram of the mapping of real-time positions of nine markers from an arbitrary located coordinate system (x_1 , x_2) into an isoparametric coordinate system (r , s).

tives. However, the received ‘normal strains’ ($\partial u_1/\partial r$, $\partial u_2/\partial s$) and the ‘measures of shear’ ($\partial u_1/\partial s$, $\partial u_2/\partial r$) have to be transformed back to the original coordinate system. The partial derivatives with respect to the reference coordinates (X_1 , X_2) are multiplied by the so-called Jacobian matrix $[\mathbf{J}]$. Hence, we receive the partial derivatives with respect to the parameters r and s according to

$$\begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{\partial}{\partial s} \end{bmatrix} = [\mathbf{J}] \begin{bmatrix} \frac{\partial}{\partial X_1} \\ \frac{\partial}{\partial X_2} \end{bmatrix}, \quad [\mathbf{J}] = \begin{bmatrix} \frac{\partial X_1}{\partial r} & \frac{\partial X_2}{\partial r} \\ \frac{\partial X_1}{\partial s} & \frac{\partial X_2}{\partial s} \end{bmatrix}. \quad (4)$$

By inverting the Jacobian matrix $[\mathbf{J}]$ under the condition that $\det[\mathbf{J}] \neq 0$, ‘normal strains’ ($\partial u_1/\partial X_1$, $\partial u_2/\partial X_2$) and the ‘measures of shear’ ($\gamma_1 = \partial u_1/\partial X_2$, $\gamma_2 = \partial u_2/\partial X_1$) can be described as a function of the parameters r and s , i.e.

$$\begin{bmatrix} \frac{\partial u_1(r, s)}{\partial \mathbf{X}} \\ \frac{\partial u_2(r, s)}{\partial \mathbf{X}} \end{bmatrix} = \begin{bmatrix} \frac{\partial u_1(r, s)}{\partial X_1} \\ \frac{\partial u_1(r, s)}{\partial X_2} \end{bmatrix} = [\mathbf{J}^{-1}] \begin{bmatrix} \frac{\partial u_1}{\partial r} \\ \frac{\partial u_1}{\partial s} \end{bmatrix}, \quad (5)$$

$$\begin{bmatrix} \frac{\partial u_2(r, s)}{\partial \mathbf{X}} \\ \frac{\partial u_2(r, s)}{\partial \mathbf{X}} \end{bmatrix} = \begin{bmatrix} \frac{\partial u_2(r, s)}{\partial X_1} \\ \frac{\partial u_2(r, s)}{\partial X_2} \end{bmatrix} = [\mathbf{J}^{-1}] \begin{bmatrix} \frac{\partial u_2}{\partial r} \\ \frac{\partial u_2}{\partial s} \end{bmatrix}.$$

Myocardial tissue is inhomogeneous and orthotropic due to the structure and the different constituents equipped with different nonlinear mechanical properties. Nevertheless, we assume a planar homogeneous biaxial deformation $\chi(\mathbf{X})$, which is described as

$$x_1 = \lambda_1 X_1 + \gamma_1 X_2, \quad x_2 = \gamma_2 X_1 + \lambda_2 X_2, \quad x_3 = \lambda_3 X_3, \quad (6)$$

where λ_i denote stretch ratios along the axes of the orthonormal system, with $i = \{1, 2, 3\}$, and γ_1 and γ_2 are ‘measures of the in-plane shear’, as mathematically derived above. The deformation gradient $\mathbf{F}$ is then given in the matrix form by

$$[\mathbf{F}] = \left[\frac{\partial \chi(\mathbf{X})}{\partial \mathbf{X}} \right] = \begin{bmatrix} \lambda_1 & \gamma_1 & 0 \\ \gamma_2 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}. \quad (7)$$

Based on the high water content of the human ventricular myocardium and the low permeability, the myocardial tissue can be assumed as an incompressible material.⁴¹ With this assumption, the stretch λ_3 in the third direction (throughout the thickness of the tissue) can then be determined as¹²

$$J = \det[\mathbf{F}] = \lambda_3(\lambda_1\lambda_2 - \gamma_1\gamma_2) \stackrel{!}{=} 1, \quad (8)$$

$$\Rightarrow \lambda_3 = (\lambda_1\lambda_2 - \gamma_1\gamma_2)^{-1}.$$

By means of the introduced isoparametric interpolation, we are now able to provide the deformation gradient at an arbitrary point within the tracked region. Subsequently, this deformation gradient serves as the basis for the calculations of the normal and in-plane shear stresses within the tested specimen. A homogenous deformation gradient from the nine marker positions was calculated.

Kinetics—Measure of Stress

In this section, mathematical relations for the determination of components of the Cauchy stress matrix, i.e., normal and shear stresses as a function of the specimen size in the reference configuration (length L_1 , width L_2 , thickness T), stretch ratios (λ_i) and ‘measures of shear’ (γ_i , $i = \{1, 2\}$), and the recorded forces (f_1, f_2)

are stated. Note, we are using a convention for the components σ_{ij} of the Cauchy stress matrix $\boldsymbol{\sigma}$ according to Truesdell and Noll,³⁹ where the first index i denotes the direction $\mathbf{e}_i$ in which the stress acts, and where the second index j indicates that the stress acts on a plane normal to $\mathbf{e}_j$; compare with Fig. 5a.

Humphrey *et al.*¹⁷ performed biaxial extension tests on the epicardium excised from ventricular free walls of canine hearts. They quantified the 2D deformation during the test and calculated Cauchy stresses from the first Piola-Kirchhoff stresses and the deformation gradient. By assuming that P_{12} and P_{21} are approximately zero, this first approach does not ensure the symmetry $[\boldsymbol{\sigma}] = [\boldsymbol{\sigma}^T]$, because P_{11} , P_{22} , κ_1 , and κ_2 act independently. Hence, there is a need to improve this approach.

By (theoretically) considering forces in a biaxial extension test caused by the clamping and/or the fiber orientation of the specimen, the frequently used first Piola-Kirchhoff stress matrix $[\mathbf{P}]$ can be determined. This way, the actual force is referred to the cross sectional area in the reference configuration so that the related stress matrix is written as

$$[\mathbf{P}] = \begin{bmatrix} \frac{f_{11}}{L_2 T} & \frac{f_{12}}{L_1 T} & \frac{f_{13}}{L_1 L_2} \\ \frac{f_{21}}{L_2 T} & \frac{f_{22}}{L_1 T} & \frac{f_{23}}{L_1 L_2} \\ \frac{f_{31}}{L_2 T} & \frac{f_{32}}{L_1 T} & \frac{f_{33}}{L_1 L_2} \end{bmatrix}. \quad (9)$$

Since we are investigating the mechanical behavior of thin specimens, in the (1, 2)-plane, the thin sheet or ‘membrane’ approximation allows us to set $P_{13} = P_{23} = P_{33} = P_{31} = P_{32} = 0$.¹³ Therefore, the related first Piola-Kirchhoff stress matrix gives

$$[\mathbf{P}] = \begin{bmatrix} \frac{f_{11}}{L_2 T} & \frac{f_{12}}{L_1 T} & 0 \\ \frac{f_{21}}{L_2 T} & \frac{f_{22}}{L_1 T} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (10)$$

This statement only holds if f_{13} , f_{23} , f_{33} , f_{31} , and f_{32} become zero. An illustration of the forces f_{11} , f_{12} , f_{21} , and f_{22} is depicted in Fig. 5b, where the coordinates of one node of the square having sides of 2 are indicated as $x_1 = \lambda_1 + \gamma_1$ and $x_2 = \lambda_2 + \gamma_2$.

Since we can only measure the sum of the forces along the directions $\mathbf{e}_1$ and $\mathbf{e}_2$, i.e., f_1 and f_2 , we now have four unknown variables: f_{11} , f_{12} , f_{21} , f_{22} , and need four equations to solve this problem. We get two equations from the previously described force measurement, i.e.

$$f_1 = f_{11} + f_{12}, \quad f_2 = f_{21} + f_{22}. \quad (11)$$

By applying the Piola transformation, we finally receive the Cauchy stress matrix as (note that we have here used $J = 1$)

$$[\boldsymbol{\sigma}] = [\mathbf{P}][\mathbf{F}]^T = \begin{bmatrix} \lambda_1 P_{11} + \gamma_1 P_{12} & \lambda_2 P_{12} + \gamma_2 P_{11} & 0 \\ \lambda_1 P_{21} + \gamma_1 P_{22} & \lambda_2 P_{22} + \gamma_2 P_{21} & 0 \\ 0 & 0 & 0 \end{bmatrix} = [\boldsymbol{\sigma}^T]. \quad (12)$$

As $[\boldsymbol{\sigma}]$ is a symmetric matrix, we can write $[\boldsymbol{\sigma}] = [\boldsymbol{\sigma}^T]$ so that

$$\sigma_{12} = \sigma_{21}, \quad (13)$$

which, together with Eq. (12), provides the third equation.

The fourth equation can be obtained from the relation between the Cauchy stress matrix $[\boldsymbol{\sigma}]$ and the Lagrange multiplier p . We equate the Lagrange multiplier of the Cauchy stress matrix, which Humphrey *et al.*²¹ derived in their study, with the Lagrange multiplier of the Cauchy stress matrix we derived, as Freed *et al.*⁸ utilized in their study, and obtain

$$p = -\frac{1}{3} \text{tr}[\boldsymbol{\sigma}] = -\frac{1}{3}(\sigma_{11} + \sigma_{22}) \stackrel{!}{=} -\frac{1}{3} \left(\frac{\lambda_1 f_1}{L_2 T} + \frac{\lambda_2 f_2}{L_1 T} \right), \quad (14)$$

which, together with Eq. (10) in (12), gives the fourth equation.

Hence, the linear system of equations has the following form:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \gamma_2 L_1 & \lambda_2 L_2 & -\lambda_1 L_1 & -\gamma_1 L_2 \\ \lambda_1 L_1 & \gamma_1 L_2 & \gamma_2 L_1 & \lambda_2 L_2 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{12} \\ f_{21} \\ f_{22} \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ 0 \\ \lambda_1 L_1 f_1 + \lambda_2 L_2 f_2 \end{bmatrix}. \quad (15)$$

By inverting the left matrix, as long as the determinant is unequal to zero, the unknown forces f_{11} , f_{12} , f_{21} , and f_{22} can be computed. Using Eqs. (10) and (12), we then obtain the required relations between measured forces and the recorded deformations, and we are able to plot in-plane stresses as a function of in-plane deformations.

To quantify differences between the two testing modes, we introduce a ‘measure of anisotropy’ ξ , which describes the degree of anisotropy of a biaxially tested material. Thus,

$$\xi = \frac{|\sigma_{11} - \sigma_{22}|}{\max(\sigma_{11}, \sigma_{22})} 100\%. \quad (16)$$

For an isotropic state $\xi = 0$.

In order to validate the obtained results, the (theoretical) fiber direction can be directly calculated from the determined normal and shear Cauchy stresses by using Mohr’s circle for the two-dimensional state of stress. Therefore, we can calculate the fiber angle between the basis vector $\mathbf{e}_1$ and the mean fiber direction $\mathbf{e}_f$ by

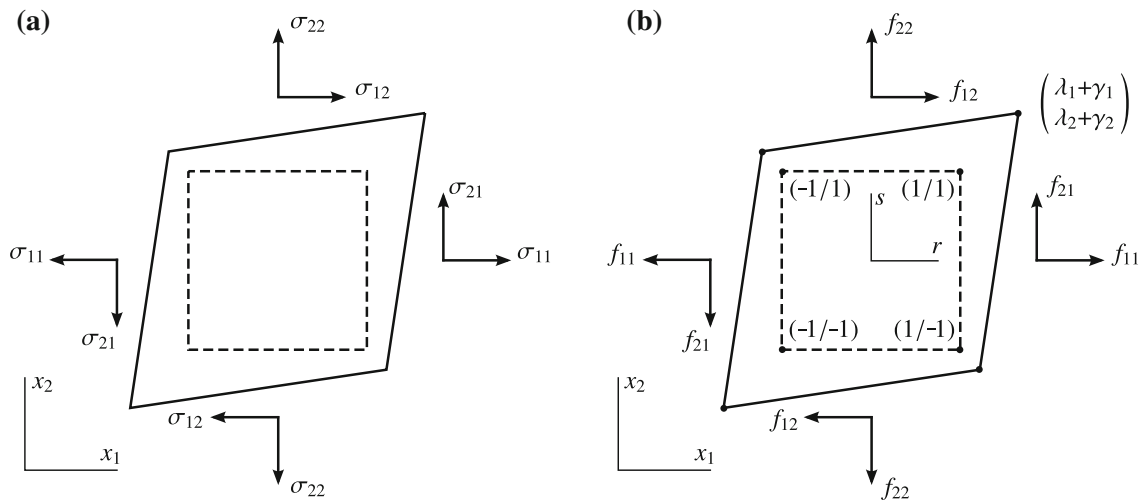


FIGURE 5. Schematic diagram showing normal and shear stresses in (a) and associated forces in (b), which occur in a biaxial extension test. Stress and force notations are adapted from Ogden.²⁶ Dashed lines indicate the referential (unloaded) configuration, where solid lines represent the deformed (loaded) configuration. In addition, the isoparametric coordinate system with related coordinates is depicted.

$$\beta_{\text{calc}} = \frac{1}{2} \arctan \left(\frac{2\sigma_{12}}{\sigma_{11} - \sigma_{22}} \right). \quad (17)$$

RESULTS

Load transmission—Latex material

The effects of the clamping method and, therefore, the transmission of forces from the setup to the specimen were evaluated by means of equibiaxial extension tests on an isotropic elastic natural rubber material (Latex). As depicted in Fig. 6a, normal Cauchy stresses have (very) similar maximum values in both the MFD and the cross-MFD, which reflects the expected isotropic behavior of the tested Latex material. Furthermore, it showed almost linear behavior up to 2% extension, whereby at higher stretches it became a nonlinear stiffening material. Corresponding shear Cauchy stresses vs. the ‘measure of shear’ are plotted in Fig. 6b. For this material, the ‘measure of shear’ γ_1 and γ_2 are identical and, therefore, labeled as γ . The shear Cauchy stress vs. ‘measure of shear’ relationship was pronounced nonlinear, initially very flat, and became very steep at higher values of γ . By comparing the calculated maximum shear Cauchy stresses σ_{12} with normal Cauchy stresses σ_{11} , σ_{22} (see Table 1), the percentage of the ratios of the shear Cauchy stress to the normal Cauchy stress ($\sigma_{12}:\sigma_{11}$ and $\sigma_{12}:\sigma_{22}$) were determined to be smaller than 0.2%, therefore, very small. Moreover, these ratios were relatively independent of the applied stretches. The ‘measure of anisotropy’ ξ and the calculated fiber angle β_{calc} were small

($\approx 4\text{--}6\%$ and $\approx 2\text{--}3^\circ$, respectively) revealing the (almost) isotropic character of the tested Latex material. All the above stated results indicate a well-performing force transmission between setup and specimens.

Human LV myocardium

Representative preconditioning behavior of the human LV myocardium at 10% equibiaxial stretch and testing mode 1 ($\beta = 0^\circ$) is shown in Fig. 7. After only three cycles, no differences between the loading and unloading curves of the 3rd and 4th cycle in the stress-stretch plot were visible. The same qualitative preconditioning behavior was obtained for the myocardium in testing mode 2 ($\beta = 45^\circ$) (not shown herein).

Figure 8a shows typical measurement cycles (normal Cauchy stress vs. stretch) obtained from equibiaxial extension tests on specimen II at different stretches (5–15% in 2.5% increments) and in testing mode 1 ($\beta = 0^\circ$). Corresponding shear Cauchy stresses vs. ‘measures of shear’ are plotted in Fig. 8b. Note, that values of the ‘measure of shear’ γ_1 and γ_2 vary significantly from each other. For this reason, we decided to show both, σ_{12} vs. γ_1 and σ_{12} vs. γ_2 . Since all experiments performed on the LV yield qualitatively similar results, we abstained from demonstrating the normal and shear Cauchy stress curves for every specimen. In general, the human myocardium tested in mode 1 revealed highly nonlinear and anisotropic material behavior with stiffer behavior in the MFD than in the cross-MFD. In contrast, the LV myocardium tested in mode 2 showed isotropic behavior,

as expected, for a fiber-reinforced material with a fiber angle of $\beta = 45^\circ$. A pronounced hysteresis between loading and unloading curves was observed for both testing modes, indicating high dissipative properties of the myocardium during cyclic loading. Moreover, the passive LV myocardium showed softening behavior with increasing stretches. This softening seemed to be more pronounced in testing mode 1 than that of 2, and also more severe in MFD than in cross-MFD.

Table 1 lists the percentages of the stress ratios ($\sigma_{12}:\sigma_{11}$ and $\sigma_{12}:\sigma_{22}$) of equibiaxially tested representative specimen *II* in mode 1 and specimen *IX* tested in mode 2 for different stretch levels. The percental stress ratios for testing mode 1 were small ($\leq 1\%$) until the stretch level of 12.5%. In contrast, specimen *IX* tested in mode 2 revealed higher stress ratios ranging from 2.4 to 6.2% for the same stretch levels. Interestingly, at the highest applied stretch (15%), the percental stress ratio in specimen *II* tested in mode 1 increased to 5.2%, indicating changes in the fiber orientation or architecture due to possible tissue damage. For both testing modes, the stress ratios increased with increasing stretch. For specimen *II* tested in mode 1, the ‘measure of anisotropy’ values ξ were determined to be high ($> 48\%$). The ξ -values decreased from 68 to 48% with increased extension, indicating some changes in the myocytes orientation. In contrast, the specimen *IX* tested in mode 2 showed very small ξ -values ($< 3\%$), suggesting an isotropic material when fibers in the myocardium are mainly oriented in 45° to the loading directions. The calculated fiber angles β_{calc} were close to 0° for testing mode 1, and between 32° and 45° for testing mode 2. This conforms the well-working preparation technique used to obtain specimens with fibers angles $\beta = 0^\circ$ and $\beta = 45^\circ$.

Although the myocardial behavior is qualitatively well represented in Figs. 8a and 8c (normal Cauchy stresses), and Figs. 8b and 8d (shear Cauchy stresses), some variations between the different specimens have been observed. The column plot in Fig. 9 depicts the peak normal Cauchy stresses in the MFD and cross-MFD as well as corresponding shear Cauchy stresses of equibiaxial extension tests on all tested myocardial specimens at a stretch level of 7.5%. The mechanical behavior of all specimens was qualitatively similar, but quantitatively varied; we observed varying maximum Cauchy stresses from specimen to specimen. At an equibiaxial stretch of 7.5%, the mean values for both percental stress ratios ($\sigma_{12}:\sigma_{11}$ and $\sigma_{12}:\sigma_{22}$) for all specimens tested in mode 1 ($n = 7$) were determined to be $0.7 \pm 0.3\%$ and $1.4 \pm 0.6\%$, respectively, with a ‘measure of anisotropy’ $\xi = 45.3 \pm 12.7\%$ and a calculated fiber angle $\beta_{calc} = 1.0 \pm 0.5^\circ$. Corresponding values for all specimens tested in mode 2 ($n = 7$) were

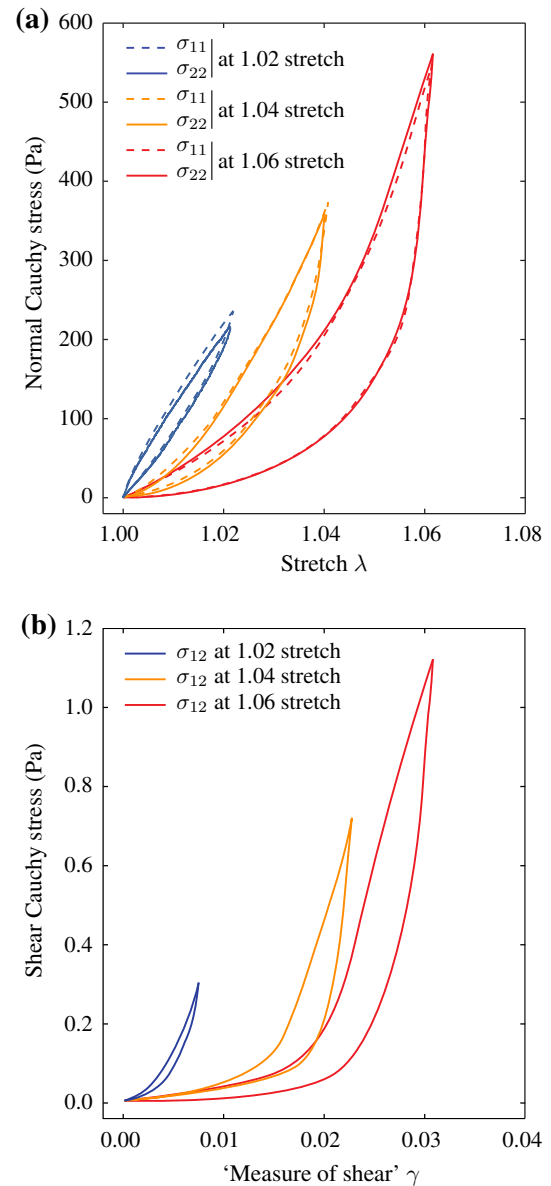


FIGURE 6. Measurement cycles (loading and unloading) of an equibiaxial extension test of an isotropic elastic material (Latex) at different stretches: (a) normal Cauchy stresses (σ_{11} , σ_{22}) in the coordinate axes x_1 and x_2 are plotted vs. stretches; (b) in-plane shear Cauchy stresses (σ_{12}) are plotted vs. the ‘measure of shear’ γ ; $\gamma = \gamma_1 = \gamma_2$ for the (Latex) material.

$$\sigma_{12}:\sigma_{11} = 12.5 \pm 13.1\%, \quad \sigma_{12}:\sigma_{22} = 12.8 \pm 13.5\%, \\ \xi = 2.3 \pm 1.7\%, \quad \text{and } \beta_{calc} = 41.7 \pm 6.7^\circ.$$

DISCUSSION

Diastolic mechanical properties of cardiac muscle are important determinants of the cardiac function with distinct passive myocardial stiffness contributing to diastolic heart failure.¹¹ Biaxial extension tests on

TABLE 1. ‘Measures of shear’ (γ_1, γ_2), normal (σ_{11}, σ_{22}) and shear Cauchy stresses (σ_{12}), corresponding percentage of shear to normal Cauchy stress ratios ($\sigma_{12} : \sigma_{11}, \sigma_{12} : \sigma_{22}$), then the measure of anisotropy (ξ) and the fiber angle (β_{calc}) calculated from Mohr’s circle for the two-dimensional state of stress at different stretch ratios for the biaxially tested elastic, isotropic Latex material, representative human myocardium specimen *II* in testing mode 1 and specimen *IX* in testing mode 2.

Stretch ratios	Measures of shear		Cauchy stresses			Stress ratios		Measure of anisotropy	Calc. fiber angle
$\lambda_1 = \lambda_2$ (–)	γ_1 (10^{-2})	γ_2 (10^{-2})	σ_{11} (kPa)	σ_{22} (kPa)	σ_{12} (Pa)	$\sigma_{12} : \sigma_{11}$ (%)	$\sigma_{12} : \sigma_{22}$ (%)	ξ (%)	β_{calc} (°)
Elastic, isotropic Latex material									
1.02	0.7	0.7	0.2	0.2	0.4	0.2	0.2	4.3	2.0
1.04	2.2	2.2	0.4	0.3	0.8	0.2	0.2	5.9	2.0
1.06	3.1	3.1	0.6	0.6	1.1	0.2	0.2	3.8	2.9
Human myocardium specimen <i>II</i> (testing mode 1: $\beta = 0^\circ$)									
1.050	0.4	0.9	7.0	2.2	16.9	0.3	0.8	68.2	0.2
1.075	0.9	1.0	9.9	3.8	25.2	0.3	0.7	61.9	0.2
1.100	1.1	0.9	13.2	6.1	41.8	0.3	0.7	53.9	0.3
1.125	2.2	1.5	19.1	9.5	92.2	0.5	1.0	50.3	0.6
1.150	4.3	6.7	28.7	15.0	776.7	2.7	5.2	47.6	3.2
Human myocardium specimen <i>IX</i> (testing mode 2: $\beta = 45^\circ$)									
1.050	0.7	6.0	1.2	1.2	30.2	2.4	2.5	2.4	31.8
1.075	1.5	9.0	3.4	3.4	143.9	4.2	4.2	0.3	44.0
1.100	2.5	12.2	7.1	7.0	366.5	5.3	5.2	2.0	39.6
1.125	3.5	12.0	10.3	10.0	639.6	6.2	5.4	2.0	40.7
1.150	2.4	13.5	16.8	16.3	1000.8	6.0	6.1	3.0	37.9

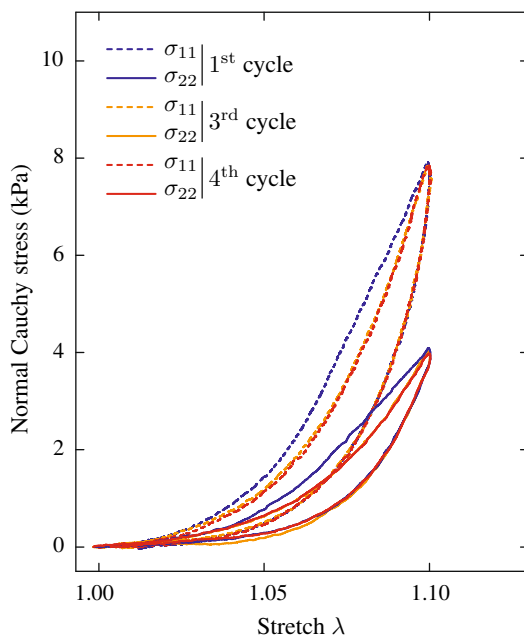


FIGURE 7. Preconditioning behavior of equibiaxially tested left ventricular human myocardium at 10% stretch: Four preconditioning cycles of one representative specimen are shown. Cauchy stress σ_{11} is associated with the mean fiber direction, where σ_{22} is the stress in the cross fiber direction. After the third preconditioning cycle, no changes of the loading and unloading curves can be observed.

the passive myocardium mimic loading conditions during the diastolic filling phase. Moreover, biaxial extension tests of fiber-reinforced soft biological tissues have been found as an appropriate technique to characterize the (anisotropic) tissue mechanics and to de-

velop constitutive formulations, because it better represents multiaxial tissue *in vivo* deformations than uniaxial testing.

Unfortunately, during biaxial extension tests of soft biological tissues, shear deformations may occur and cannot be controlled during testing. It is interesting to note that most published studies neglect shear stresses by saying that they are small. Hence, it is important to check whether shear stresses exist because they would definitely influence normal stresses and, therefore, the material parameters to be obtained. Thus, the focus of this (preliminary) study lied on the determination of the shear stresses in biaxially tested human myocardial specimens, where on the one hand the fibers were oriented towards one loading axis (minimizing the shear) and on the other hand the fibers were rotated 45° regarding the loading axis (maximizing the shear).

To the authors’ knowledge, this is the first study focusing on the determination of normal and shear stresses in human myocardial tissue obtained from biaxial extension tests. Inspired by the works of Humphrey and colleagues,^{17–20,22,23} we were able to improve the strain quantification they introduced by increasing the number of detected markers and by using shape functions, which raise the accuracy of the finite element based mapping method and the strain calculation. In addition, we used the same approach as Freed *et al.*⁸ to calculate a Cauchy stress matrix, which has to be symmetric.

One main finding of the present study is that in-plane shear stresses definitely exist in biaxially tested human myocardium, but can be reduced to a minimum

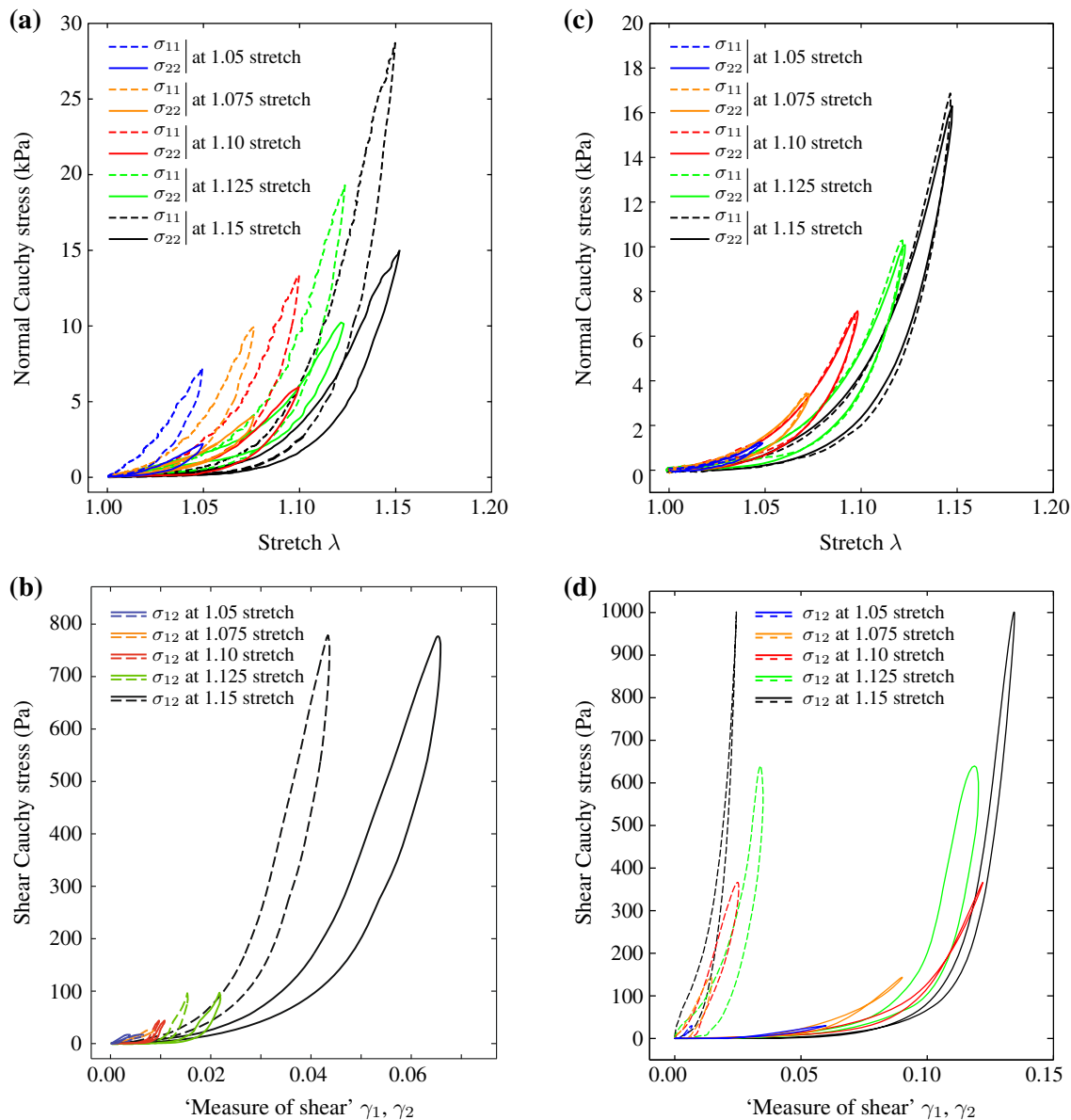


FIGURE 8. Representative equibiaxial measurement cycles of human myocardial specimen *II* tested in mode 1 (a)–(b) and human myocardial specimen *IX* tested in mode 2 (c)–(d) at different stretch levels. Panels (a) and (c) show normal Cauchy stresses σ_{11} and σ_{22} in the directions $\mathbf{e}_1$ (dashed curves) and $\mathbf{e}_2$ (solid curves), respectively, where panels (b) and (d) show shear Cauchy stresses vs. ‘measure of shear’: γ_1 (solid curves), γ_2 (dashed curves). Note, for specimen *II*, $\mathbf{e}_1$ was aligned with MFD, and $\mathbf{e}_2$ with cross-MFD. For specimen *IX*, the angle β between $\mathbf{e}_1$ and the MFD was 45° .

by an effective preparation of the specimens. Optimal soft tissue specimens for biaxial extension tests have to be squared, cut as thin as possible, arranged so that the MFD is aligned with one side, and hooked with at least five equidistantly placed hooks on each side for an appropriate load transfer and insertion into the testing setup.⁴ Furthermore, by means of an elastic isotropic material (Latex) we could show that the force transmission between the testing device and the specimen by means of hooks and cords does not influence our experiments and, therefore, does not falsify the results.

In general, the passive myocardium revealed non-linear stiffening and direction-dependent behavior which is typically the case for fiber-reinforced soft biological tissue.^{9,16,32–34} In particular, the myocardial tissue showed pronounced anisotropic behavior when tested with a fiber angle of $\beta = 0^\circ$, and showed almost isotropic behavior when tested with a fiber angle of $\beta = 45^\circ$. Additionally, the tissue exhibited distinct hysteresis formation during cyclic loading and unloading, which has been observed for muscular tissues.³⁸ According to Humphrey,¹⁶ the hysteresis is

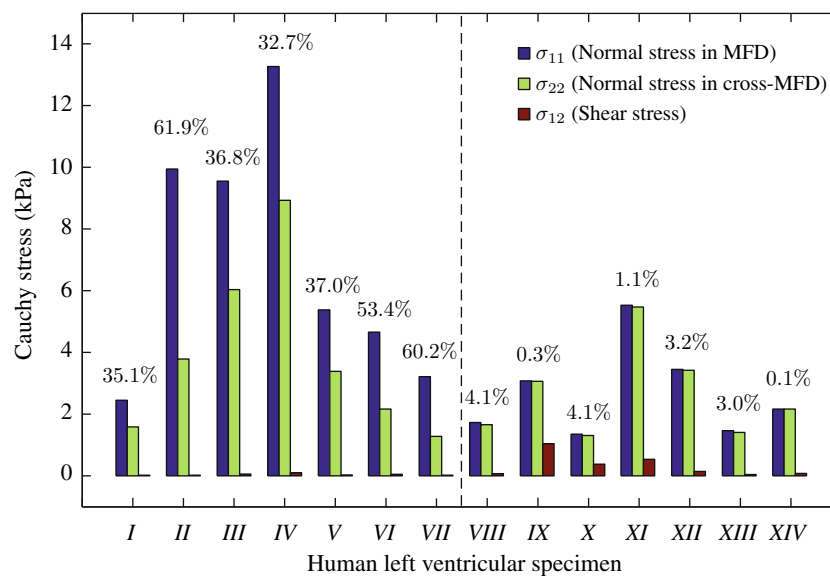


FIGURE 9. Comparison of in-plane normal and shear Cauchy stresses of equibiaxially tested 14 myocardial specimens at an applied stretch of $\lambda = 1.075$. For specimens I–VII, the MFD was aligned along $\mathbf{e}_1$ (testing mode 1), while for specimens VIII–XIV, the angle β between $\mathbf{e}_1$ and the MFD was 45° (testing mode 2). Values for the measure of anisotropy ξ , i.e., Eq. (16), are shown above the bars for each specimen. Mean values and standard deviations of ξ were $45.3 \pm 12.7\%$ for testing mode 1 and $2.3 \pm 1.7\%$ for testing mode 2. Calculated mean fiber angles β_{calc} of $1.0 \pm 0.5^\circ$ and $41.7 \pm 6.7^\circ$ were obtained for testing mode 1 and 2, respectively.

consistent with a viscoelastic response, which is related to the substantial water content (about 80% by wet weight) but also the presence of muscle. Moreover, we observed softening with increasing stretch, indicating irreversible damage of the muscle cells/fibers during biaxial loadings. These observations are not discussed in detail, since this was not the scope of this study. For a more detailed discussion about the properties of the passive human ventricular myocardium, the reader is referred to a follow-up study by Sommer *et al.*³⁵

The method we used appears to be robust to quantify normal and shear deformations and stresses in biaxially tested human myocardium, even though some assumptions have been made. Since the in-plane orientation of muscle fibers changes continuously through the thickness of the sample,²⁷ ‘out of plane’ shearing will occur during biaxial extension tests but can not be measured. Here we prepared specimens as thinly as possible to neglect this effect, but the thickness could not be reduced to less than ≈ 2 mm because cutting soft biological tissues always causes difficulties such as the insertion of damage. Due to the relatively large ratio of the specimen’s side length to thickness of $> 10:1$, the thin sheet approximation can be assumed.

In conclusion, we state that shear stresses definitely exist in biaxially tested human myocardium, but are very small compared to normal stresses in the MFD and the cross-MFD (see Fig. 9) for appropriately prepared test specimens. The preparation procedures and the methods used to quantify normal and shear deformations and stresses were robust and easily

applicable to the human ventricular myocardium and, therefore, suggested to be generally used in biaxial extension tests of fiber-reinforced soft biological tissues. The above stated methodologies and findings are utilized in a subsequent biomechanical study on the human myocardium.³⁵ In that study, biaxial extension tests, triaxial shear tests, and microstructural imaging were performed on different regions of the human ventricular myocardium. The combination of biaxial extension and triaxial shear test data with the underlying microstructure will lead to material parameters for an appropriate description of the mechanical properties of the passive human myocardium.^{14,36} By coupling of the passive mechanical properties based on that data and with the knowledge of myocardial electrophysiology, more realistic numerical heart simulations could be performed providing a better understanding of the fundamental mechanisms of complex mechanics in the heart wall. This is a step needed towards the improvement of medical treatment of heart diseases.

ACKNOWLEDGMENTS

The authors are grateful to A. Abbasi for her substantial contributions to the experiments, and to F. Heinzl from Medical University of Graz, Department of Cardiology, for many discussions on this subject matter. We thank also Daniel Han for his editorial support during preparing this manuscript. This project

was supported by the Austrian Science Fund (FWF) with Grant number P 23830-N13. The authors gratefully acknowledge this support.

REFERENCES

- ¹Chew, P. H., F. C. Yin, and S. L. Zeger. Biaxial stress-strain properties of canine pericardium. *J. Mol. Cell Cardiol.* 18:567–578, 1986.
- ²Demer, L. L., and F. C. P. Yin. Passive biaxial mechanical properties of isolated canine myocardium. *J. Physiol. Lond.* 1983;339:615–630.
- ³Dokos, S., B. H. Smaill, A. A. Young, and I. J. LeGrice. Shear properties of passive ventricular myocardium. *Am. J. Physiol.* 283:H2650–H2659, 2002.
- ⁴Eilaghi, A., J. G. Flanagan, G. W. Brodland, and C. R. Ethier. Strain uniformity in biaxial specimens is highly sensitive to attachment details. *J. Biomech. Eng.* 131:0910031–0910037, 2009.
- ⁵Eriksson, T. S. E., A. J. Prassl, G. Plank, and G. A. Holzapfel. Influence of myocardial fiber/sheet orientations on left ventricular mechanical contraction. *Math. Mech. Solids* 18:592–606, 2013.
- ⁶Eriksson, T. S. E., A. J. Prassl, G. Plank, and G. A. Holzapfel. Modeling the dispersion in electro-mechanically coupled myocardium. *Int. J. Numer. Methods Biomed. Eng.* 29:1267–1284, 2013.
- ⁷Frank, J. S., and G. A. Langer. The myocardial interstitium: its structure and its role in ionic exchange. *J. Cell Biol.* 60:586–601, 1974.
- ⁸Freed, A. D., D. R. Einstein, and M. S. Sacks. Hypoelastic soft tissues: part II: in-plane biaxial experiments. *Acta Mech.* 2010;213:205–222.
- ⁹Fung, Y. C. Biomechanics. Mechanical Properties of Living Tissues, 2nd ed. New York: Springer-Verlag, 1993.
- ¹⁰Ghaemi, H., K. Behdinan, and A. D. Spence. In vitro technique in estimation of passive mechanical properties of bovine heart part I. Experimental techniques and data. *Med. Eng. Phys.* 31:76–82, 2009.
- ¹¹Gutierrez, C., and D. G. Blanchard. Diastolic heart failure: challenges of diagnosis and treatment. *Am. Family Phys.* 69:2609–2616, 2004.
- ¹²Holzapfel, G. A. Nonlinear Solid Mechanics. A Continuum Approach for Engineering. Chichester: John Wiley & Sons, 2000.
- ¹³Holzapfel, G. A., and R. W. Ogden. On planar biaxial tests for anisotropic nonlinearly elastic solids. A continuum mechanical framework. *Math. Mech. Solids* 14:474–489, 2009.
- ¹⁴Holzapfel, G. A., R. W. editors. Biomechanical Modelling at the Molecular, Cellular and Tissue Levels. Wien-New York: Springer-Verlag, 2009.
- ¹⁵Holzapfel, G. A., and R. W. Ogden. Constitutive modelling of passive myocardium: a structurally based framework for material characterization. *Philos. Trans. R. Soc. A.* 367:3445–3475, 2009.
- ¹⁶Humphrey, J. D. Cardiovascular Solid Mechanics. Cells, Tissues, and Organs. New York: Springer-Verlag, 2002.
- ¹⁷Humphrey, J. D., D. L. Vawter, and R. P. Vito. Quantification of strains in biaxially tested soft tissues. *J. Biomech.* 20:59–65, 1987.
- ¹⁸Humphrey, J. D., and F. C. P. Yin. On constitutive relations and finite deformations of passive cardiac tissue—Part I: a pseudo-strain energy function. *J. Biomech. Eng.* 109:298–304, 1987.
- ¹⁹Humphrey, J. D., and F. C. Yin. Biomechanical experiments on excised myocardium: theoretical considerations. *J. Biomech.* 22:377–383, 1989.
- ²⁰Humphrey, J. D., and F. C. P. Yin. Constitutive relations and finite deformations of passive cardiac tissue: II. Stress analysis in the left ventricle. *Circ. Res.* 65:805–817, 1989.
- ²¹Humphrey, J. D., R. K. Strumpf, and F. C. P. Yin. Biaxial mechanical behavior of excised ventricular epicardium. *Am. J. Physiol. Heart Circ. Physiol.* 259:H101–H108, 1990.
- ²²Humphrey, J. D., R. K. Strumpf, and F. C. P. Yin. Determination of constitutive relation for passive myocardium: I. A new functional form. *J. Biomech. Eng.* 112:333–339, 1990.
- ²³Humphrey, J. D., R. K. Strumpf, and F. C. P. Yin. Determination of constitutive relation for passive myocardium: II. Parameter estimation. *J. Biomech. Eng.* 112:340–346, 1990.
- ²⁴Langdon, S. E., R. Chernecky, C. A. Pereira, and D. A. J. M. Lee. Biaxial mechanical/structural effects of equibiaxial strain during crosslinking of bovine pericardial xenograft materials. *Biomaterials* 20:137–153, 1999.
- ²⁵Nielsen, P. M. F., I. J. LeGrice, B. H. Smaill, and P. J. Hunter. Mathematical model of geometry and fibrous structure of the heart. *Am. J. Physiol. Cell. Physiol.* 260:H1365–H1378, 1991.
- ²⁶Ogden, R. W. Non-linear Elastic Deformations. New York: Dover, 1997.
- ²⁷Rohmer, D., A. Sitek, and G. T. Gullberg. Reconstruction and visualization of fiber and laminar structure in the normal human heart from ex vivo diffusion tensor magnetic resonance imaging (DTMRI) data. *Investig. Radiol.* 42:777–789, 2007.
- ²⁸Sands, G. B., B. H. Smaill, and I. J. LeGrice. Virtual sectioning of cardiac tissue relative to fiber orientation. In: Proceedings of the 30th Annual International Conference of the IEEE Engineering in Medicine and Biology Society EMBS 2008, 2008, pp. 226–229.
- ²⁹Sands, G. B., D. A. Gerneke, D. A. Hooks, C. R. Green, B. H. Smaill, and I. J. LeGrice. Automated imaging of extended tissue volumes using confocal microscopy. *Microsc. Res. Tech.* 67:227–239, 2005.
- ³⁰Scollan, D. F., A. Holmes, R. Winslow, and J. Forder. Histological validation of myocardial microstructure obtained from diffusion tensor magnetic resonance imaging. *Am. J. Physiol. Heart. Circ. Physiol.* 275(6):H2308–H2318, 1998.
- ³¹Sharpe, W. N. Jr., J. Pulskamp, D. G. Gianola, C. Eberl, R. G. Polcawich, and R. J. Thompson. Strain measurement of silicon dioxide microspecimens by digital image processing. *Exp. Mech.* 47:649–658, 2007.
- ³²Sommer, G., P. Regitnig, L. Köllringer, and G. A. Holzapfel. Biaxial mechanical properties of intact and layer-dissected human carotid arteries at physiological and supra-physiological loadings. *Am. J. Physiol. Heart Circ. Physiol.* 298:H898–H912, 2010.
- ³³Sommer, G., M. Eder, L. Kovacs, H. Pathak, L. Bonitz, C. Mueller, et al. Multiaxial mechanical properties and constitutive modeling of human adipose tissue: a basis for preoperative simulations in plastic and reconstructive surgery. *Acta Biomater.* 9:9036–9048, 2013.
- ³⁴Sommer, G., A. Schriefl, G. Zeindlinger, A. Katzensteiner, H. Ainödhofer, A. Saxena, et al. Multiaxial mechanical response and constitutive modeling of esophageal tissues: impact on esophageal tissue engineering. *Acta Biomater.* 9:9379–9091, 2013.

- ³⁵Sommer, G., A. J. Schriebl, M. Andrä, M. Sacherer, C. Viertler, H. Wolinski, *et al.* Biomechanical properties and microstructure of human ventricular myocardium. *Acta Biomater.* (submitted).
- ³⁶Sommer, G., M. Schwarz, M. Kutschera, R. Kresnik, P. Regitnig, A. J. Schriebl, *et al.* Biomechanical properties of the human ventricular myocardium. *Biomedizinische Technik.* 58(Suppl. 1), 2013.
- ³⁷Streeter, D. D., and W. T. Hanna. Engineering mechanics for successive states in canine left ventricular myocardium. I. Cavity and wall geometry. *Circ. Res.* 33:639–655, 1973.
- ³⁸Toursel, T., L. Stevens, H. Granzier, and Y. Mounier. Passive tension of rat skeletal soleus muscle fibers: effects of unloading conditions. *J. Appl. Physiol.* 92:1465–1472, 2002.
- ³⁹Truesdell, C., and W. Noll. In: *The Non-linear Field Theories of Mechanics*, 3rd edn, edited by S. S. Antman. Berlin: Springer-Verlag; 2004.
- ⁴⁰Yin, F. C. P. Ventricular wall stress. *Circ. Res.* 49:829–842, 1981.
- ⁴¹Yin, F. C. P., C. Chan, and R. Judd. Compressibility of perfused passive myocardium. *Am. J. Physiol. Heart Circ. Physiol.* 8:1864–1870, 1996.
- ⁴²Young, A. A., I. J. Legrice, M. A. Young, and B. H. Smaill. Extended confocal microscopy of myocardial laminae and collagen network. *J. Microsc.* 192:139–150, 1998.
- ⁴³Zienkiewicz, O. C., and R. L. Taylor. *The Finite Element Method. The Basis*, vol. 1, 5th ed. Oxford: Butterworth Heinemann, 2000.